[7] MCU - IOs

MCU 1

U6B

STM32G474RET6

I/O

SWDIO1	49	PA13	SWD	PORT A	PA0 12 ECA_T1
SWCLK1	50	PA14	SWD	PORT A	PA1 13 ECB_T1
NRST1	7	PG10	SWD	PORT A	PA2 14 OCM_T1
				PORT A	PA3 17 OCM_T2
				PORT A	PA4 18 ECC_T2
				PORT A	PA5 19 ECC_T1
OSC_IN_1	5	PF0	PORT F	PORT A	PA6 20 ECB_T2
OSC_OUT_1	6	PF1	PORT F	PORT A	PA7 21 ECA_T2
				PORT A	PA8 42 I2C_SDA
				PORT A	PA9 43 I2C_SCL
EN_T1	55	PD2	PORT D	PORT A	PA10 44 GPIO_LED_Debug_1
				PORT A	PA11 45 USB1_N
				PORT A	PA12 46 USB1_P
				PORT A	PA15 51 GPIO_LED_T2
PWMA_T1	8	PC0	PORT C	PORT B	PB0 24 OCM_I
PWMB_T1	9	PC1	PORT C	PORT B	PB1 25 EN_T2
PWMA_T2	10	PC2	PORT C	PORT B	PB2 26 BOARD_VERSION
PWMB_T2	11	PC3	PORT C	PORT B	PB3 56
ECC_I	22	PC4	PORT C	PORT B	PB4 57
SENSE_VBAT	23	PC5	PORT C	PORT B	PB5 58
ECA_I	38	PC6	PORT C	PORT B	PB6 59
ECB_I	39	PC7	PORT C	PORT B	PB7 60
I2C_SCL_1	40	PC8	PORT C	PORT B	PB8 61 BOOT_1
I2C_SDA_1	41	PC9	PORT C	PORT B	PB9 62 PWM_BUZZER
GPIO_LED_T1	52	PC10	PORT C	PORT B	PB10 30 EN_I
GPIO_LED_I	53	PC11	PORT C	PORT B	PB11 33 VBUS_DET1
MCU_ID1	54	PC12	PORT C	PORT B	PB12 34 FDCAN1_RX
DIAG_T1	2	PC13	PORT C	PORT B	PB13 35 FDCAN1_TX
DIAG_T2	3	PC14	PORT C	PORT B	PB14 36 PWMB_I
DIAG_I	4	PC15	PORT C	PORT B	PB15 37 PWMA_I

MCU 2

U7B

STM32G474RET6

I/O

SWDIO2	49	PA13	SWD	PORT A	PA0 12 ECB_M
SWCLK2	50	PA14	SWD	PORT A	PA1 13 ECA_M
NRST2	7	PG10	SWD	PORT A	PA2 14 OCM_M
				PORT A	PA3 17 OCM_R
				PORT A	PA4 18 ECC_M
				PORT A	PA5 19 ECC_R
OSC_IN_2	5	PF0	PORT F	PORT A	PA6 20 ECB_R
OSC_OUT_2	6	PF1	PORT F	PORT A	PA7 21 ECA_R
				PORT A	PA8 42 I2C_SDA
				PORT A	PA9 43 I2C_SCL
EN_R	55	PD2	PORT D	PORT A	PA10 44 GPIO_LED_Debug_2
				PORT A	PA11 45 USB2_N
				PORT A	PA12 46 USB2_P
				PORT A	PA15 51 GPIO_LED_M
PWMA_R	8	PC0	PORT C	PORT B	PB0 24 OCM_P
PWMB_R	9	PC1	PORT C	PORT B	PB1 25 EN_M
PWMB_M	10	PC2	PORT C	PORT B	PB2 26
PWMA_M	11	PC3	PORT C	PORT B	PB3 56
ECC_P	22	PC4	PORT C	PORT B	PB4 57
SENSE_VBAT	23	PC5	PORT C	PORT B	PB5 58
ECB_P	38	PC6	PORT C	PORT B	PB6 59
ECA_P	39	PC7	PORT C	PORT B	PB7 60
I2C_SCL_2	40	PC8	PORT C	PORT B	PB8 61 BOOT_2
I2C_SDA_2	41	PC9	PORT C	PORT B	PB9 62
GPIO_LED_R	52	PC10	PORT C	PORT B	PB10 30 EN_P
GPIO_LED_P	53	PC11	PORT C	PORT B	PB11 33 VBUS_DET2
MCU_ID2	54	PC12	PORT C	PORT B	PB12 34 FDCAN2_RX
DIAG_M	2	PC13	PORT C	PORT B	PB13 35 FDCAN2_TX
DIAG_R	3	PC14	PORT C	PORT B	PB14 36 PWMA_P
DIAG_P	4	PC15	PORT C	PORT B	PB15 37 PWMB_P

DESIGN NOTE:
We use the LQFP64 version to have extra pins for controlling an external fan.

DESIGN NOTE:
Do not forget to flash option bytes when flashing for the first time.

LAYOUT NOTE:
Footprint deviates from datasheet and IPC7351 to allow for soldermask sliver between pads according to PCBWay design rules.

Battery Voltage Sense

SENSE_VBAT → SENSE_VBAT

Board Version Selection

BOARD_VERSION → BOARD_VERSION

Auxiliary Ports

SWD (Debug) Port

SWDIO1, SWCLK1, NRST1 → {SWD1}

SWDIO2, SWCLK2, NRST2 → {SWD2}

Buzzer

PWM_BUZZER → PWM_BUZZER

MCB to MB - I2C

I2C_SDA, I2C_SCL → {I2C_CNT}

5Mbps FD-CAN Bus

FDCAN1_RX, FDCAN1_TX → {FDCAN1_TTL}

FDCAN2_RX, FDCAN2_TX → {FDCAN2_TTL}

MCU ID

MCU_ID1, MCU_ID2 → {MCU_ID}

LEDs

GPIO_LED_I, GPIO_LED_T1, GPIO_LED_T2 → {LED_Finger1}

GPIO_LED_P, GPIO_LED_R, GPIO_LED_M → {LED_Finger2}

GPIO_LED_Debug_1, GPIO_LED_Debug_2 → {LED_Debug}

Oscillator HSE

X1 — X322516MLB4SI

OSC_IN_1 — OSC1 (pin 1); OSC2 (pin 3) — OSC_OUT_1

C43 CAP CER 0603 (pin 2 → GND); C44 CAP CER 0603 (pin 4 → GND)

X2 — X322516MLB4SI

OSC_IN_2 — OSC1 (pin 1); OSC2 (pin 3) — OSC_OUT_2

C45 CAP CER 0603 (pin 2 → GND); C46 CAP CER 0603 (pin 4 → GND)

Temperature and Humidity

I2C_SDA_1, I2C_SCL_1 → {SHT40_1}

I2C_SDA_2, I2C_SCL_2 → {SHT40_2}

SWITCH

BOOT_1, BOOT_2 → {SW_BOOT}

USB PORT

USB1_P, USB1_N, VBUS_DET1 → {USB_1}

USB2_P, USB2_N, VBUS_DET2 → {USB_2}

Motor Driver

PWMA_I, PWMB_I, EN_I, DIAG_I, OCM_I, ECA_I, ECB_I, ECC_I → {MD_I}

PWMA_M, PWMB_M, EN_M, DIAG_M, OCM_M, ECA_M, ECB_M, ECC_M → {MD_M}

PWMA_R, PWMB_R, EN_R, DIAG_R, OCM_R, ECA_R, ECB_R, ECC_R → {MD_R}

PWMA_P, PWMB_P, EN_P, DIAG_P, OCM_P, ECA_P, ECB_P, ECC_P → {MD_P}

PWMA_T1, PWMB_T1, EN_T1, DIAG_T1, OCM_T1, ECA_T1, ECB_T1, ECC_T1 → {MD_T1}

PWMA_T2, PWMB_T2, EN_T2, DIAG_T2, OCM_T2, ECA_T2, ECB_T2, ECC_T2 → {MD_T2}

Test Points

Battery Voltage Sense
SENSE_VBAT — TP8

Board Version Selection
BOARD_VERSION — TP12

SWD (Debug) Port
NRST2 — TP16
SWCLK2 — TP18
SWDIO2 — TP20
NRST1 — TP21
SWCLK1 — TP23
SWDIO1 — TP25

5Mbps FD-CAN Bus
FDCAN1_TX — TP28
FDCAN1_RX — TP30
FDCAN2_TX — TP33
FDCAN2_RX — TP35

LEDs
GPIO_LED_I — TP37
GPIO_LED_T1 — TP39
GPIO_LED_T2 — TP41
GPIO_LED_P — TP43
GPIO_LED_R — TP45
GPIO_LED_M — TP47
GPIO_LED_Debug_1 — TP49
GPIO_LED_Debug_2 — TP51

Temperature and Humidity
I2C_SDA_1 — TP53
I2C_SCL_1 — TP55
I2C_SDA_2 — TP57
I2C_SCL_2 — TP59

USB PORT
USB1_P — TP63
USB1_N — TP65
VBUS_DET1 — TP66
USB2_P — TP68
USB2_N — TP70
VBUS_DET2 — TP72

MCB to MB - I2C
I2C_SCL — TP76
I2C_SDA — TP78

Buzzer
PWM_BUZZER — TP80

MCU ID
MCU_ID1 — TP85
MCU_ID2 — TP87

MD Index
PWMA_I — TP9
PWMB_I — TP10
EN_I — TP11
DIAG_I — TP13
OCM_I — TP14
ECA_I — TP15
ECB_I — TP17
ECC_I — TP19

MD Middle
PWMA_M — TP22
PWMB_M — TP24
EN_M — TP26
DIAG_M — TP27
OCM_M — TP29
ECA_M — TP31
ECB_M — TP32
ECC_M — TP34

MD Ring
PWMA_R — TP36
PWMB_R — TP38
EN_R — TP40
DIAG_R — TP42
OCM_R — TP44
ECA_R — TP46
ECB_R — TP48
ECC_R — TP50

MD Pinky
PWMA_P — TP52
PWMB_P — TP54
EN_P — TP56
DIAG_P — TP58
OCM_P — TP60
ECA_P — TP61
ECB_P — TP62
ECC_P — TP64

MD Thumb 1
PWMA_T1 — TP67
PWMB_T1 — TP69
EN_T1 — TP71
DIAG_T1 — TP73
OCM_T1 — TP74
ECA_T1 — TP75
ECB_T1 — TP77
ECC_T1 — TP79

MD Thumb 2
PWMA_T2 — TP81
PWMB_T2 — TP82
EN_T2 — TP83
DIAG_T2 — TP84
OCM_T2 — TP86
ECA_T2 — TP88
ECB_T2 — TP89
ECC_T2 — TP90

Comments:	Company: Willas (威倫電子股份有限公司 Willas Electronic Corp.)	Variant: DRAFT	Git Hash: cc04374
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Sheet Title:	File Name: MCU - IOs.kicad_sch	Designer: Damon	Date: 2026-06-27 / Revision: + (Unreleased)
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